# Rassah (رصّة): AI Packing Assistant. Project Report

**Project 1: AI Product Development and Monitoring** · Tuwaiq Academy × Startupbootcamp × Meta · 22 October 2026
**Team:** Abdulaziz Alkhlaiwe, Hana, Bayan. Lanes: AI & Data, Engine & UI, Platform & Monitoring
**Date:** 2026-10-22
**Live demo:** https://rassah.vercel.app · **Repository:** https://github.com/justAbdulaziz10/Rassah

> **Rassah is an MVP built in two days.** The brief asked each team to come up with an AI product, build it end to end between 20 and 22 October 2026, deploy it so anyone can try it, and monitor it. It runs on synthetic data, with no login and no live store integration (see section 11, Limitations).
>
> Every number comes from `eval/results/*.json`, `docs/report/results.md`, `docs/demo-script.md` or the dashboard.

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## 1. Problem

Merchants who ship multi-item orders choose cartons by eye. Three things go wrong:

1. **Oversized cartons raise shipping cost.** Couriers bill the larger of actual and volumetric weight (`L × W × H / 5000`).
2. **Fragile items break and liquids leak.** A new packer puts a book on a perfume bottle, or lays an oil bottle flat.
3. **The data that would fix this doesn't exist.** Professional cartonization tools assume clean product dimensions in a warehouse system. Small Saudi merchants only have free-text product pages in Arabic and English.

## 2. Solution and user flow

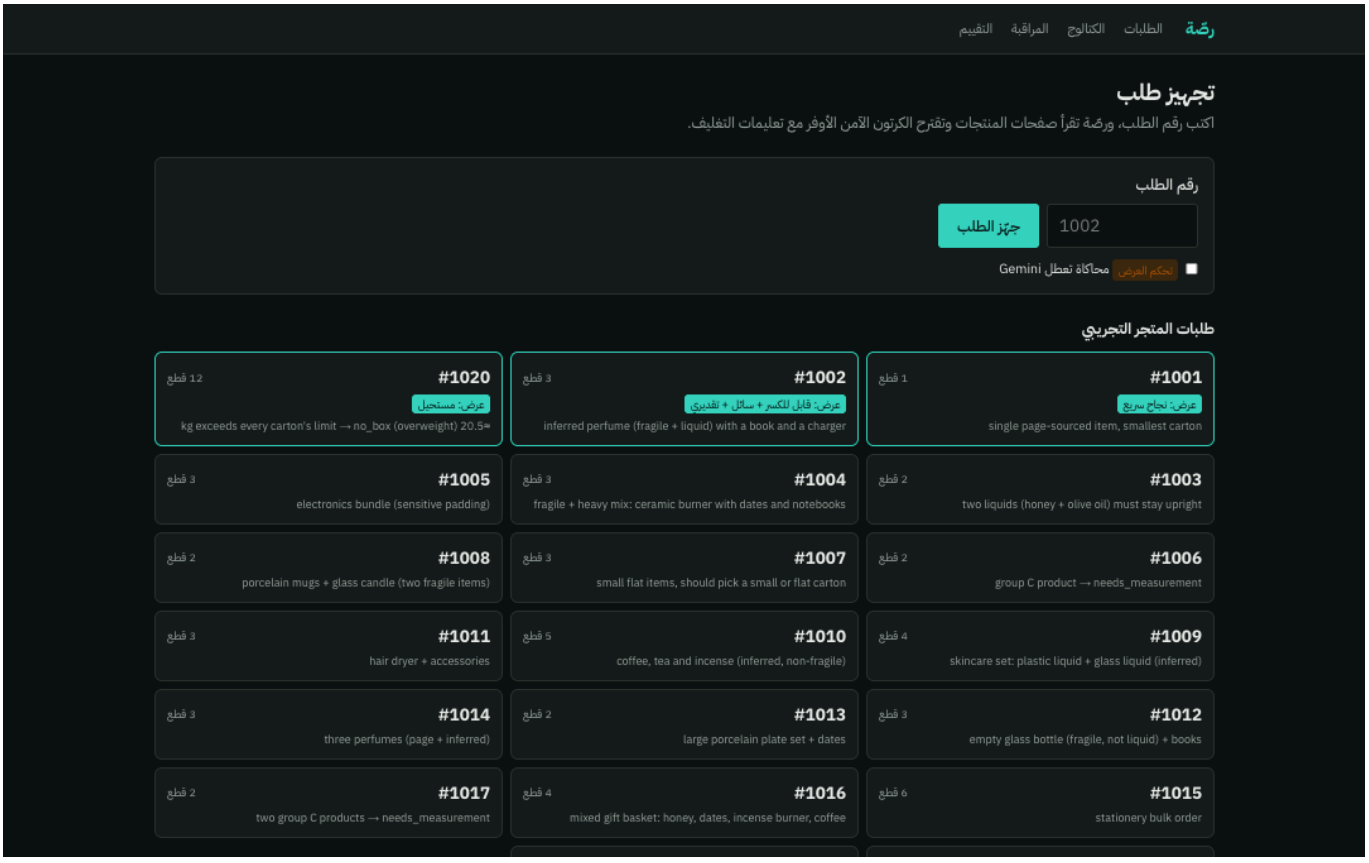
`/` is a landing page (its hero replays a real engine result for order 1016). The working screens are `/orders`, `/notifications`, `/catalog`, `/monitoring` and `/evaluation`. `/store` is a built-in demo store on the same domain and database: an order placed there appears in the `/orders` queue.

The packer types an order number. Rassah:

1. Loads the order's products. For each product not already cached, **Gemini reads the product page** and returns outer-package dimensions, weight, fragile / liquid / sensitive flags, where each value came from (*page*, *inferred*, *unknown*), confidence, and an evidence quote.
2. Runs a **constrained 3D packing search** over the carton catalog: gravity (≥ 70% base support), nothing on fragile items, liquids upright, weight limits, a safety margin on inferred sizes, and padding for fragile/sensitive items.
3. Gates every layout through an **independent validator**, then picks the cheapest valid carton by estimated carton + shipping cost.

4. Shows the carton, **numbered Arabic packing instructions**, a 3D step-by-step view, estimated cost, and per-product source badges. The packer **confirms or corrects**. Corrections are stored and measured.

If a value has no basis, the order stops with "needs measurement" (the packer measures once). If no single carton fits, it says so and suggests a split into several cartons, each one checked like a normal order. It never invents a layout.



رصّة الطلبات الكتالوج المراقبة التقييم

← كل الطلبات

## الطلب #1002

تم اختيار كرتون

# كرتون كبير L

40 × 30 × 25 سم

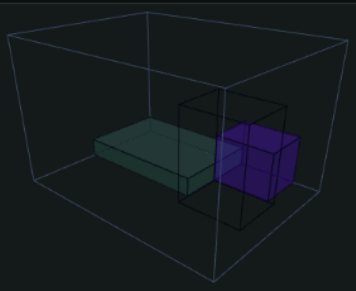
3 من الكاش · 0 استُخرج بالـ AI · 232 ms

إعادة التجهيز | تأكيد التغليف

### تعليمات التغليف

1. استخدم كرتون «كبير L» (25 × 30 × 40 سم من الداخل).
2. ضع «رواية ساق البامبو» في قاع الكرتون.
   - ⚠ مقاسه تقديري: تأكد إنه يدخل قبل إغلاق الكرتون.
3. ضع «شاحن جداري سريع 20W» في قاع الكرتون.
   - حساس: أضف حشوة حماية 1.5 سم من كل الجهات.
   - ⚠ مقاسه تقديري: تأكد إنه يدخل قبل إغلاق الكرتون.
4. ضع «عطر أو دو بارفان 50 مل» في قاع الكرتون.
   - قابل للكسر: لفّه بغلاف فقاعي (1.5 سم على الأقل) ولا تضع فوقه أي شيء.
   - سائل: أبقِه واقفاً (هذا الجانب للأعلى).
   - ⚠ مقاسه تقديري: تأكد إنه يدخل قبل إغلاق الكرتون.
5. املأ الفراغات بمواد حشو حتى لا تتحرك القطع، ثم أغلق الكرتون.
6. اكتب على الكرتون: «هذا الجانب للأعلى ↑».

### العرض ثلاثي الأبعاد

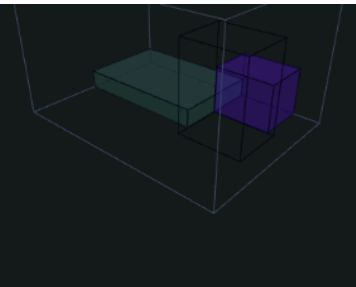


الخطوة 3 / 3

قابل للكسر · سائل · حساس · عادي · الشفاف = مقاس تقديري

### التكلفة التقديرية (أسعار تجريبية، ليست أسعار شركة شحن)

| الكرتون | الوزن الفعلي | الوزن الحجمي | الشحن | الإجمالي |
|---|---|---|---|---|
| 2.60 ر.س | 1.25 كجم | 6.61 كجم | 36.00 ر.س | **38.60 ر.س** |

الخطوة 3 / 3

قابل للكسر · سائل · حساس · عادي · الشفاف = مقاس تقديري

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|---|---|---|---|---|
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### المنتجات

| المنتج | الكمية | الأبعاد | الوزن | التعامل | المصدر | |
|---|---|---|---|---|---|---|
| عطر أو دو بارفان 50 مل<br>«inferred from 50 ml perfume bottle in a carton box» | 1 | 8 × 6 × 12 سم<br>للترتيب: 16.8 × 12.2 × 9.9 سم (+15% هامش) (+حماية) | 300 جم | قابل للكسر · سائل | تقديري | تعديل |
| رواية ساق البامبو<br>«inferred from 400 page paperback novel» | 1 | 14 × 3 × 21 سم<br>للترتيب: 24.1 × 16.1 × 3.5 سم (+15% هامش) | 400 جم | عادي | تقديري | تعديل |
| شاحن جداري سريع 20W<br>«inferred from 20W wall charger» | 1 | 5 × 5 × 7 سم<br>للترتيب: 11.1 × 8.8 × 8.8 سم (+15% هامش) (+حماية) | 100 جم | حساس | تقديري | تعديل |

رقم التشغيل: 2562a14a-e3ba-4cab-86fa-70f15668ef92

## 3. Where the AI is, and why

| Step | Technique | Why |
|---|---|---|
| Understand product pages | **LLM (Gemini)** with a JSON schema and honesty rules | Free-form Arabic/English text, mixed units, Arabic-Indic digits, implicit fragility ("كريستال") defeat rules |
| Arrange items | **Search** (classical AI): corner-grid candidates × rotations × 9 fixed orderings/scores, then 40 seeded restarts, then a backtracking anchor search | An LLM cannot guarantee collision-free, supported coordinates |
| Prove validity | Independent validator (plain geometry) | Correctness checked by code that shares nothing with the search |

"*The LLM understands, the engine computes, the validator proves.*" The instructor confirmed that using AI APIs meets the brief and that no fine-tuning is needed.

## 4. Data collection

- **Synthetic, stated openly.** 30 products written by the team in the style of Saudi store pages (no real merchant data):
  - **Group A (15):** dimensions and weight stated, in varied formats (Arabic-Indic digits, inches/lb, "الطول/العرض/الارتفاع", buried mid-paragraph).
  - **Group B (12):** clues only (50 ml, 400 pages, 12 pieces).
  - **Group C (3):** no usable information.
- **Ground truth** per product (outer box, natural upright orientation, flags, expected source). Independent review by a member other than the description author is still open (CHECKLIST P3.12), so the ground truth is AI-drafted and unreviewed. **10-product holdout** is excluded from prompt tuning.
- **20 orders** (incl. 3 demo orders, needs-measurement and impossible cases), **5 cartons**, synthetic shipping rates.
- **60 packing fixtures** from a seeded generator: 20 known-feasible (guillotine cuts of a carton, so a witness layout exists), 20 impossible (oversize / overweight / liquid too tall), 20 random.
- All data files are schema-validated in CI (`npm run validate-data`).

## 5. Technical design

Mermaid sources are in `docs/diagrams.md`. PNG exports:

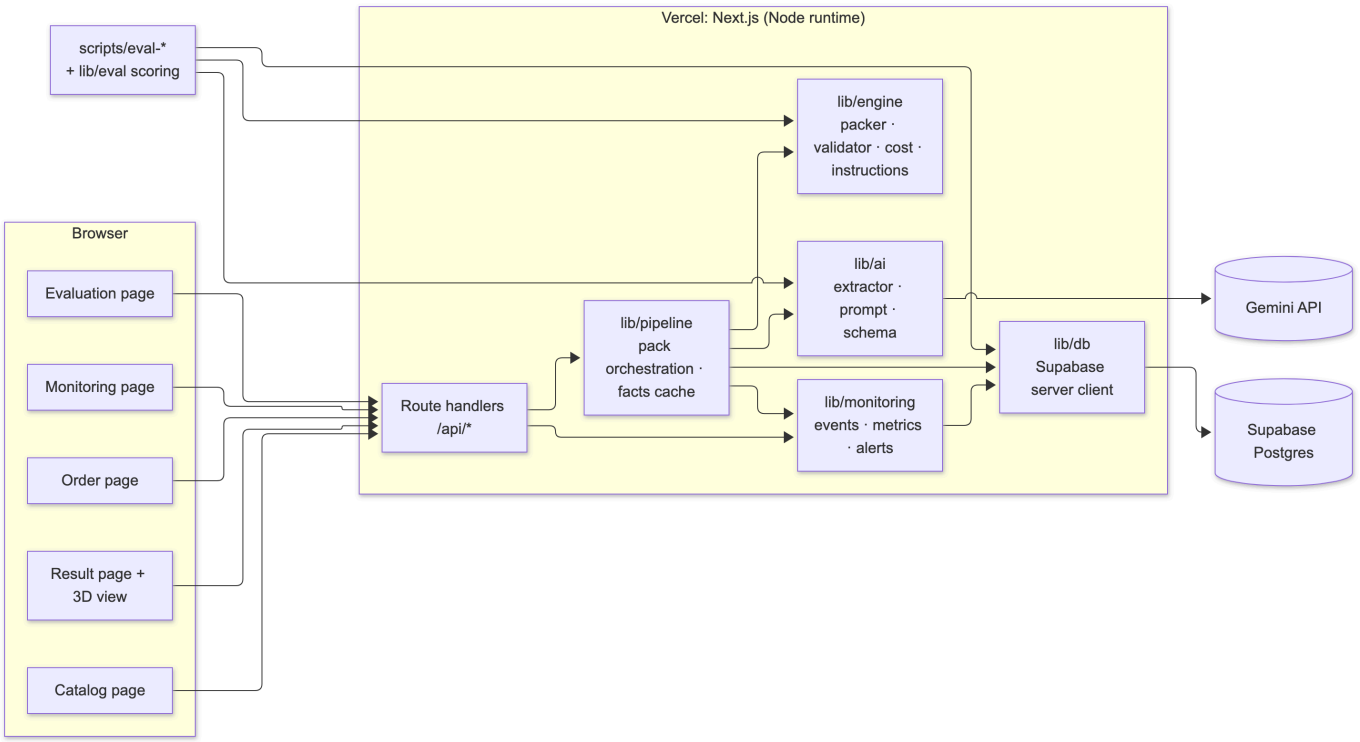



*Figure 1. System architecture: pages → route handlers → pipeline → AI / engine / monitoring → Supabase.*

- **Stack:** Next.js 16 (App Router, Cache Components) + TypeScript strict on Vercel. Supabase Postgres. `@google/genai`. react-three-fiber. Recharts. Vitest.
- **Pack pipeline** (`lib/pipeline/pack.ts`): order → catalog cache (source-hash freshness; manual/corrected facts never overwritten) → Gemini for misses (concurrency 2, 20 s timeout, retries only on 429/5xx, SDK retries off) → engine → validator → `order_runs` + `extraction_events` → alert refresh. The result is returned even if monitoring writes fail (flagged "monitoring degraded").
- **Engine** (`lib/engine/`): pure TypeScript, no I/O, deterministic. Every rule exists twice: in the search, and in the independent validator.
- **Cost model:** carton + shipping on chargeable weight. Illustrative rates, always labeled "estimated".
- **Security:** service-role key server-only, RLS on all tables with no public policies, Zod on every input, demo controls only with `DEMO_MODE=true`, product text delimited as data in the prompt.
- **Offline fallback:** `DATA_BACKEND=memory` runs the whole app without Supabase (local only).

## 6. Development environment

Agents work in git worktrees → local merge → **local CI gate** (`agent-ship ci`: secret scan, floor guard, lint, typecheck, tests, build, packing eval) → push straight to `main` → **GitHub Actions CI** (lint, typecheck with `next typegen`, data validation, 227 tests, build, packing eval) → Vercel production deploy → smoke test → dashboard. There was no per-PR review and no branch protection; per-PR previews were planned but only one preview check was done (2026-10-21). Decisions are logged with reasons in `docs/decisions.md`. Progress is tracked in `docs/CHECKLIST.md` (13 phases, gates). Exact dependency versions are pinned, and Node 24 LTS is set via `.nvmrc`.

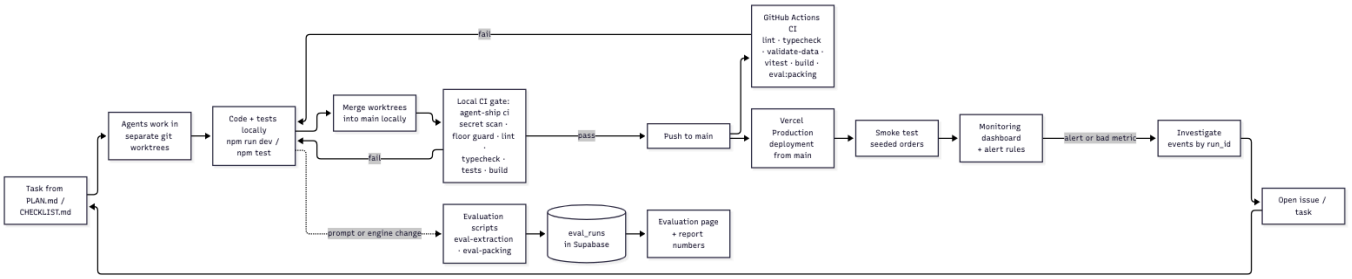



*Figure 2. Development flow as planned (branch → PR → preview → review → merge). The process actually used is the paragraph above and `docs/decisions.md`; the figure was not redrawn.*

## 7. Selected metrics (monitoring)

| # | Metric | Reveals | Alert |
|---|---|---|---|
| M1 | AI correction rate (reviewed AI products the packer corrected) | Wrong extractions | > 20% of last 30, warning |
| M2 | Data-source mix of packed units | Missing store data | > 30% unknown in 24 h, info |
| M3 | Validator rejections of engine layouts | Engine bugs (must be 0) | any in 24 h, **critical** |
| M4 | No-carton rate + skip reasons | Catalog/margin mismatch | > 15% of last 20, warning |
| M5 | p50/p95 latency, cached vs uncached | Slow Gemini vs slow engine/DB | p95 > 8 s uncached / 2 s cached over the last 20 runs, warning |
| M6 | Gemini failure rate by status | Key, quota, outage, bad output | > 10% of last 20, **critical** |
| M7 | Cost per order + cache-hit rate | Cache broken, prompt bloat | none (shown on the dashboard only) |

Every metric shows its sample size and says "insufficient data" below its minimum. Eval traffic is excluded. Alerts open once per rule, auto-resolve, and post to a webhook (never for simulated faults). Each run has a trace (Gemini calls, tokens, timings, corrections).

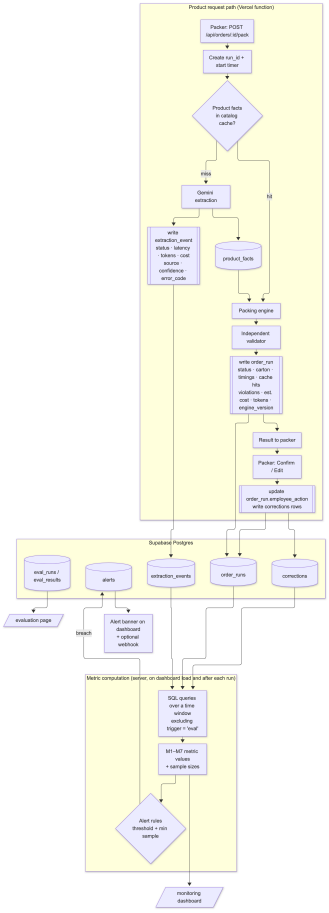



*Figure 3. Monitoring flow: events written per run, metrics M1–M7 computed over a window, alert rules.*

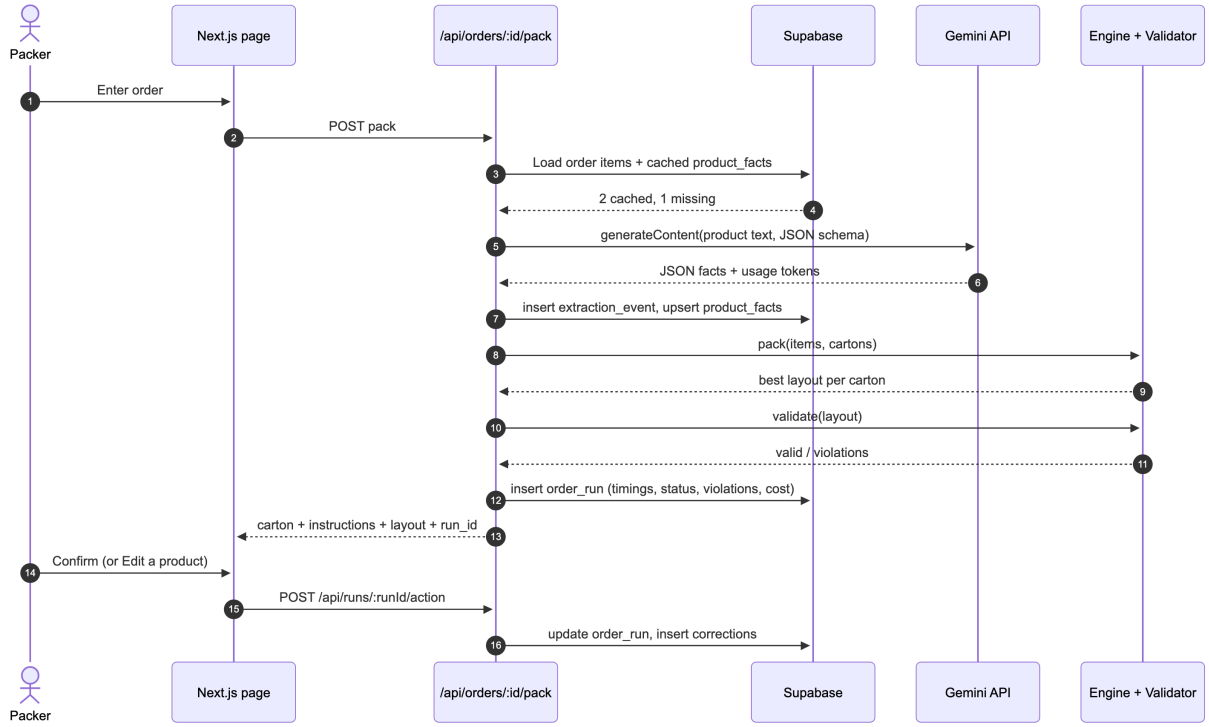



*Figure 4. One order end to end (uncached product): what is written where.*

## 8. Benchmarks and evaluation results

**Packing** (60 fixtures, `eval/results/packing-2026-10-21-engine-0.2.json`):

| Metric | Result |
|---|---|
| Engine layouts passing the independent validator | **40 / 40 (100%)** |
| Known-feasible recall (zero-slack exact fills) | **20 / 20 (100%)** |
| False claims on impossible cases | **0 / 20** |
| Naive baseline layouts failing the validator | **20 / 28**: liquid tipped 17, on fragile 7, floating 2 |
| Mean cost saving vs baseline (both valid, n = 8) | 0 SAR (**no savings claimed**) |
| Runtime p50 / p95 | 0.3 / 12.9 ms |

**Evaluation-driven iteration:** recall 3/20 (extreme points) → 9/20 (corner grid + seeded restarts; more restarts gave no gain) → 15/20 ("tight" placement scores, found by diagnosing fixture `feasible-05`) → **20/20** (engine-0.2: backtracking anchor search for the exact fills the greedy passes missed). The 0 SAR saving is by construction (both pick the cheapest fitting carton), not a bug.

**Head-to-head benchmark** (`npm run bench`, protocol in `docs/benchmark.md`, results in `eval/results/bench-2026-10-21.md`, engine-0.3). 278 cases: the 60 fixtures, 18 demo orders with true sizes and 200 seeded random orders. Every engine gets the same units, cartons and prices, and one validator and one cost model judge all of them. The harness also has adapters for two commercial packing APIs (Paccurate, 3DBinPacking); they run as soon as the team adds trial keys, so no vendor numbers are claimed here yet.

| On 258 feasible cases | Rassah | Naive packer |
|---|---|---|
| Layouts passing every rule | **100%** (98.8% one box, 1.2% split) | 32.6% |
| Cost per order where both are valid (n = 84) | **31.35 SAR** | 34.19 SAR (Rassah 8.3% cheaper, never costlier) |
| One-box-impossible fixtures (20) | 0 false claims, 7 valid splits | 6 invalid one-box layouts |
| Runtime p50 / p95 | 0.5 / 81 ms | <1 ms |

**Multi-carton split (engine-0.3):** when no single carton fits, Rassah splits the order (first-fit-decreasing by volume and by weight, then tries to empty the last box), and every box goes through the same search, validator and cost model. Demo order 1020 (≈20.5 kg) now gets a 4-carton plan with steps per carton instead of only "no carton fits".

**Extraction** (`eval/results/extraction-2026-10-21-v1.json`, `gemini-3.5-flash-lite`, prompt v1, 30 synthetic products, 30/30 calls ok). Table = run 2 (latest). Two runs were made (same model and prompt): estimated-dim MAPE 14.3% -> 12.9%, holdout 18.2% -> 16.1%, and one fragile false positive in run 2. This is LLM run-to-run variation; the committed file and the /evaluation page show run 2 (latest), run 1 is "previous".

| Metric | All (30) | Tuning (20) | Holdout (10) |
|---|---|---|---|
| Parse success | 100% | 100% | 100% |

| Metric | All (30) | Tuning (20) | Holdout (10) |
|---|---|---|---|
| Dims within 1 cm / weight within 10% (group A) | 100% / 100% | 100% / 100% | 100% / 100% |
| Inferred dim MAPE (group B) | 12.9% | 11.3% | 16.1% |
| Inferred weight MAPE (group B) | 13.5% | 11.9% | 16.7% |
| Source correctness | 100% | 100% | 100% |
| Honesty on no-info products | 100% | 100% | 100% |
| Fragile recall / precision | 100% / 90.9% | 100% / 87.5% | 100% / 100% |
| Liquid accuracy | 100% | 100% | 100% |
| Avg latency / tokens | 1795 ms / 632 | 1976 ms / 632 | 1433 ms / 633 |
| Cost per product | $0.000452 | $0.000453 | $0.000450 |

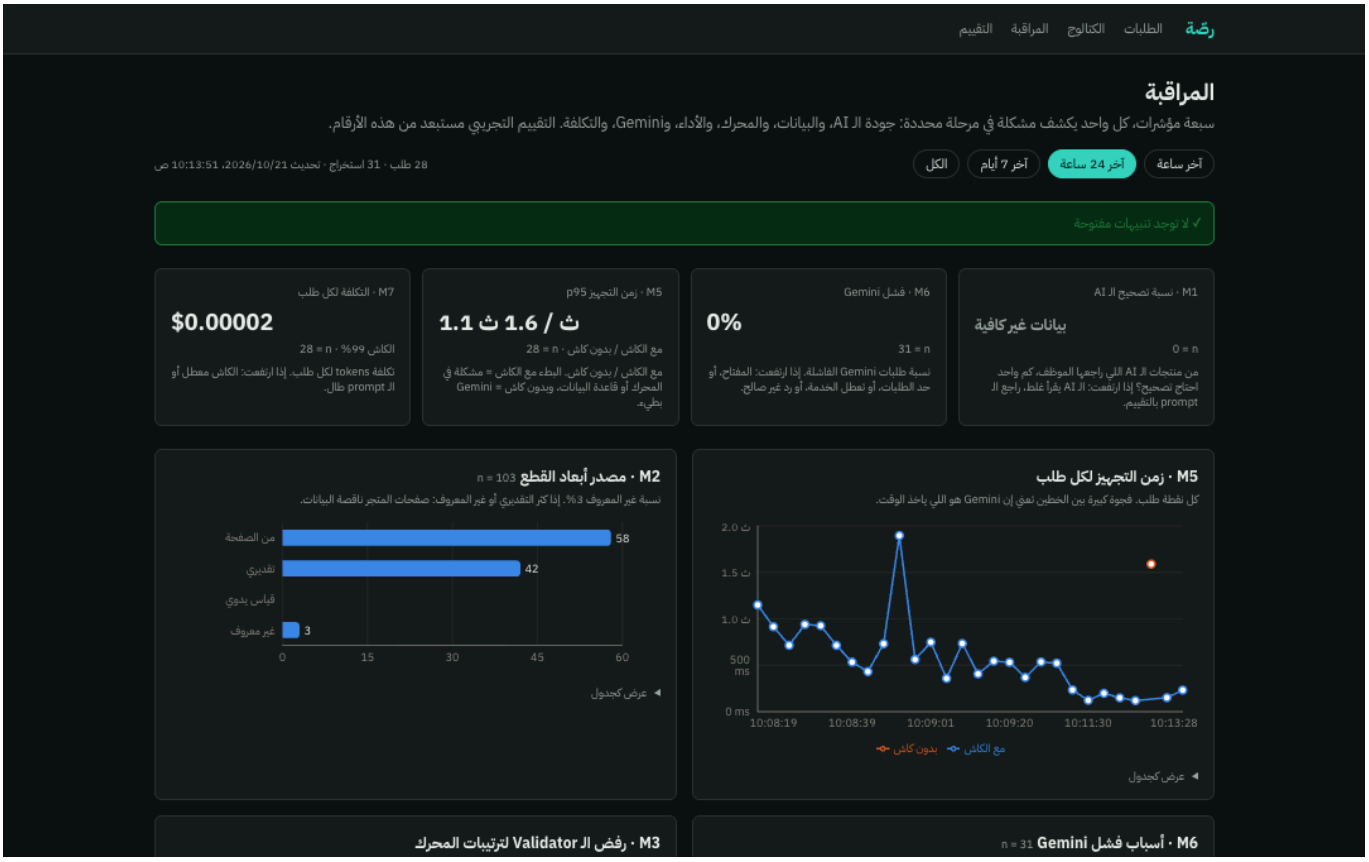
**Prompt iteration:** none. Tuning-set failures were limited to estimate error on inferred products (no extraction failures), so the prompt stays `v1` and there is no before/after row.

**Margin decision:** inferred dim MAPE over all 12 inferred products is 12.9%, rounded up to 15%, so `marginInferred` stays 0.15. The holdout MAPE (16.1%, 4 products) is above 15%. With so few inferred products this is noisy, and the instructions already say "check the fit". A real-merchant pilot should re-measure it before the margin is trusted or raised.

**Model:** `gemini-3.5-flash-lite` ($0.30 in / $2.50 out per 1M tokens, 2026-10-21). The faster-tier models had a free quota of 20 requests/day, too low for a 30-call eval.

## 9. Monitoring in action

- **Investigation 1: slow cached orders (2026-10-21).**
  - *Problem:* after the first production deploy, a smoke test of all 20 orders looked slow.
  - *Metric:* the M5 latency chart on `/monitoring` showed cached orders taking **358-1896 ms** server-side.
  - *Trace:* the `x-vercel-id` response header showed the functions running in `iad1` (US East), while Supabase is in `eu-central-1` (Frankfurt). Every DB query crossed the Atlantic.
  - *Fix:* `vercel.json` `"regions": ["fra1"]`, pushed the same day (logged in `docs/decisions.md`).
  - *After:* cached orders **119-234 ms**. One uncached order (Gemini call): **1589 ms** total, of which extraction 1205 ms. The drop is visible in the screenshot below.
- **Investigation 2: Gemini quota.** `rate_limited` events (429 QuotaFailure) showed the Flash models' free tier is 20 requests/day/model. We switched to `gemini-3.5-flash-lite` (500/day); see `docs/decisions.md`.
- **Investigation 3 (offline):** the packing eval and its fixture trace led to the scoring fix in section 8. A regression test guards it.
- **Live demo:** a simulated Gemini outage (`/orders/<id>?outage=1`, honoured only with `DEMO_MODE=true`) turns uncached products into "needs measurement" while cached ones still pack. M6 crosses 10% and opens a critical alert marked "(عطل تجريبي)". The run trace shows `simulated_outage`. A simulated alert resolves as soon as the next extraction succeeds.




## 10. Tools and AI usage

Summary of `docs/report/TOOLS.md`: Gemini (`@google/genai`) is the runtime AI feature. Claude Code (Anthropic), including its subagents on 2026-10-20/21, generated most code drafts, the docs and the synthetic data. Google AI Studio was used for the key and manual prompt trials. Everything was reviewed by the team and verified by tests, CI, evals, an end-to-end API run and screenshots. Every member walks through code they didn't write (CHECKLIST P11.44–P11.49).

## 11. Limitations

Synthetic data. Rigid box-shaped products only (no clothing). Multi-carton splits are heuristic (not the cheapest possible split). Illustrative costs. No live Salla/Zid integration (a read-only client for an external store's Orders API exists, off unless `STORE_API_URL` and `STORE_API_KEY` are set). Time-bounded heuristic (not globally optimal, although all 20/20 exact-fill fixtures are found). No crush/stacking-load or void-fill modeling. The 70% support and 1.5 cm padding are assumptions.

## 12. Next steps

Run the head-to-head benchmark against the commercial APIs with trial keys. Test on real orders (packer vs Rassah: carton spend, courier invoices, damage, packing time). Salla/Zid integration. A smarter multi-carton split. Manufacturer-spec lookup. Scale-weight feedback to catch bad facts. Separate preview/production databases.

## 13. References


- Project 1 brief (Tuwaiq Academy × Startupbootcamp × Meta), October 2026 (no URL in the repository).
- Gemini API docs: https://ai.google.dev/gemini-api/docs
- Gemini API pricing (as of 2026-10-21): https://ai.google.dev/gemini-api/docs/pricing
- Supabase docs: https://supabase.com/docs
- Vercel docs: https://vercel.com/docs
- Next.js docs: https://nextjs.org/docs (the version used is also bundled in `node_modules/next/dist/docs/`)
- React Three Fiber: https://r3f.docs.pmnd.rs
- Paccurate API documentation (existing cartonization product, used for positioning). No URL is recorded in the repository.
- py3dbp (`enzoruiz/3dbinpacking`, MIT): reference for the naive first-fit baseline idea. Our code is independent TypeScript. No URL is recorded in the repository.
- Courier volumetric-weight divisor 5000 cm³/kg: described in `docs/PLAN.md` as "commonly 5000"; set in `data/cartons.json` (`shipping.volumetricDivisor`). No courier page has been cited yet.
- Bin-packing papers: none read or cited.